Fed-PIEdgeGT A Source Holdout Pilot Study of Centralized and Federated Smart Grid Intrusion Detection

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Abstract—Smart-grid intrusion detection requires evaluation that separates genuine predictive information from benchmark-specific patterns. This paper presents Fed-PIEdgeGT v2, a reproducible source-file-holdout pipeline comparing tabular classifiers, a compact edge-aware graph transformer, and four federated optimizers on the MSU/ORNL power-system attack benchmark. Training-only preprocessing, grouped inner tuning, out-of-fold calibration, and source-aware reporting are integrated within an outer evaluation protocol. The reported experiment is a pilot using one outer fold and one training seed, with 62,635 training rows and 15,732 rows from three held-out files. The multilayer perceptron achieves the highest source macro-F1 of 0.6109, while XGBoost attains the highest pooled AUROC of 0.6841. The centralized graph model scores 0.5974 source macro-F1; FedAvg, the strongest federated variant, scores 0.5583. False-positive rates remain between 0.4709 and 0.6943 among the trained models. These findings establish a transparent benchmark and expose limitations in operating-point selection and federated optimization. They do not establish graph or physics-informed superiority. Verified physical metadata, controlled architecture ablations, and complete repeated source-holdout evaluation remain necessary before deployment claims.

Index Terms—smart-grid cybersecurity, intrusion detection, source holdout, graph transformer, federated learning, calibration.

I. Introduction

Power-system monitoring combines electrical measurements with control actions and communication logs. An intrusion detector must distinguish malicious behavior from legitimate faults, maintenance operations, and changes in system state. The MSU/ORNL power-system benchmark provides synchrophasor measurements together with simulated control-panel, relay, and Snort logs for this task [1]. Earlier work demonstrated the value of heterogeneous, time-synchronized observations for event classification and hybrid intrusion detection [2], [3].

The practical challenge is broader than fitting a high-capacity classifier to pooled rows. If records from the same acquisition group occur in both training and testing, performance can reflect group-specific signatures. Model selection, preprocessing, calibration, and threshold adjustment can introduce additional optimism when their information boundaries are unclear. Nested evaluation separates the selection stage from final assessment, although its reliability still depends on meaningful groups and adequate independent observations [4].

Graph-based learning and federated optimization are attractive extensions. Relay measurements can be arranged into node features, while attention uses connectivity and pairwise telemetry to form a structured representation. Federated learning permits local optimization across data partitions with model-update aggregation. Neither structure nor decentralization guarantees improved discrimination, and local training alone does not establish formal privacy. Their benefit must be assessed against strong centralized competitors under explicit, comparable protocols.

This study compares centralized and federated detectors through the executed Fed-PIEdgeGT v2 pilot. Its contributions are source-file-aware evaluation, matched reporting of the implemented families, and analysis of class imbalance, probability quality, and false alarms. All numerical comparisons use one configuration; earlier architectures are not pooled into this experiment.

II. Related Work

A. Power System Event Detection and Tabular Learning

Pan et al. studied classification using heterogeneous power-system data and developed a hybrid detector based on learned system behavior [2], [3]. Their work motivates combining electrical and cyber observations, but does not make every sampled benchmark row an independent utility or chronological sequence. Tabular ensembles therefore remain necessary comparators. XGBoost uses regularized tree boosting [5], while ExtraTrees randomizes splitting decisions to build an ensemble [6]. These methods test whether a graph representation adds value beyond direct feature interactions.

B. Graph Representations and Physical Knowledge

Graph Attention Networks introduce masked attention over node neighborhoods [7]. The present model uses multihead dot-product attention with edge-dependent bias and a global token; it is not an implementation of the original GAT operator. Physical knowledge can also regularize learning. Physics-informed neural networks incorporate governing equations into a training objective [8], whereas reconstructing engineered sequence components is an auxiliary representation task. These approaches should not be conflated. Physical features remain

disabled in the executed experiment because channel mapping and angle metadata have not been certified.

C. Federated Optimization and Calibration

FedAvg averages local parameters [9]; FedProx adds a broadcast-model penalty [10]; adaptive federation modifies server updates [11]; and SCAFFOLD corrects client drift [12]. Their original results motivate comparisons but do not guarantee performance on this benchmark. Temperature scaling [13] calibrates confidence; its effect must be separated from threshold-dependent classification quality.

III. Methodology

A. Dataset Validation and Label Definition

The loader requires data1.csv through data15.csv, a common 129-column schema, and the binary marker values Attack and Natural. Attack is encoded as one and Natural as zero. Here, Natural denotes the nonattack class available in the binary files; it should not be interpreted as a separate clean-operation category. Each row contains 128 predictors organized as four relay groups with 29 electrical channels and three cyber-log channels per relay.

Before splitting, predictor-identical clusters occurring across files are removed in their entirety. Remaining within-file duplicates retain one representative, and predictor-identical rows with conflicting labels are excluded. This conservative curation reduces exact-record overlap, but does not identify near-duplicate sessions or guarantee independent acquisition. Conflict exclusion also uses labels during benchmark curation and must be disclosed. The retained population in this run contains 78,367 rows; conclusions apply to that curated population.

B. Grouped Selection and Information Boundaries

A stratified grouped splitter assigns whole source files to five outer folds using split seed 42. Only fold zero is executed. Its three test files are reserved while the other twelve files supply training and inner validation. Three grouped inner folds select candidate configurations. Every inner training partition fits its own preprocessing statistics, and source-separation assertions prevent shared filenames between fitting and validation. Fig. 1 summarizes the implemented procedure.

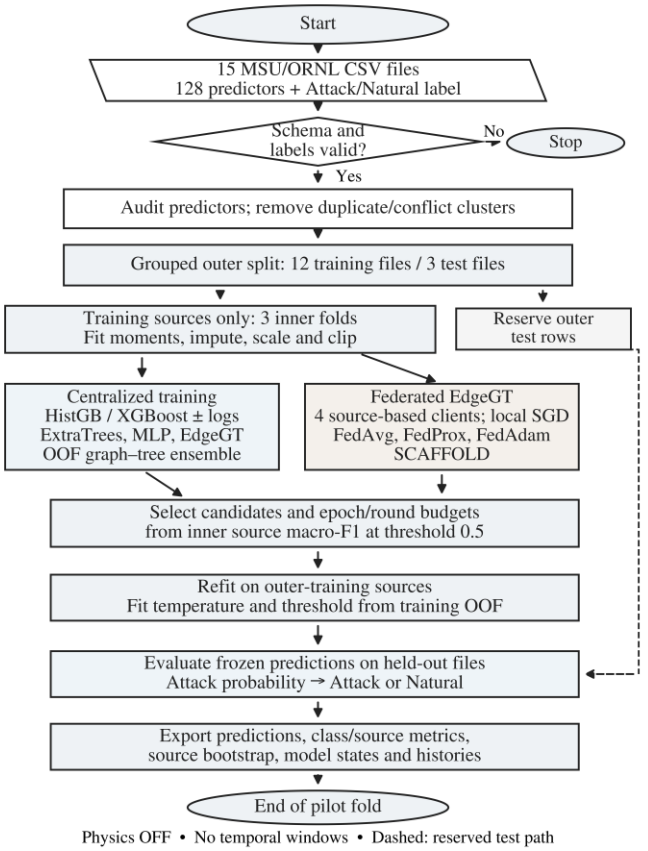


Fig. 1. Implemented Fed-PIEdgeGT v2 workflow. All tuning and probability postprocessing use outer-training sources. The dashed path reserves test rows for final evaluation. Physics features and temporal windows are inactive in the reported pilot.

Candidate ranking uses mean inner source macro-F1 at threshold 0.5. Neural early stopping uses the same criterion; the final refit budget is the median selected epoch or round count. Test labels are used only for reported outcomes. Because validation also selects checkpoints and candidates, inner out-of-fold predictions are not independent of every selection decision. The outer evaluation, rather than the selected inner score, is the relevant estimate of held-out performance.

C. Training Only Feature Preparation

Finite-value counts, sums, and squared sums determine training means and variances. Missing and infinite values are imputed with those means; standardized channels are clipped to the interval $[-8, 8]$. Missingness indicators are appended. Consequently, a relay has 64 processed channels when logs are included, and the flat representation has 256 dimensions. The no-log branch starts with 29 channels per relay and appends corresponding indicators. Channels without training variation are masked.

The same processed flat representation feeds the tree models. Thus, the raw suffix denotes inclusion of logs, not native unscaled features. Edge channels encode normalized pairwise differences in frequency, frequency deviation, one voltage magnitude, and one current magnitude. These four differences are generic telemetry features, not residuals of electrical network equations. In the federated simulator, additive client-

local moments yield equivalent training-only normalization statistics.

D. Compact Edge Aware Graph Transformer

The neural graph backbone embeds four relay nodes into 64-dimensional vectors. Learned node identities distinguish relay positions. A global token is appended, and two transformer blocks each apply four-head attention with a learned linear bias from the edge channels, residual updates, layer normalization, and a feedforward sublayer. The default mask includes self-connections, relay pairs R1–R2 and R3–R4, and connections to the global token. This mask is a configurable hypothesis, not a certified topology for independent utility networks.

A graph head predicts a logit from the mean relay embedding combined with the global token. A parallel tabular branch applies a linear layer, layer normalization, GELU activation, dropout, and an output layer to the flattened nodes. Their logits are mixed as

$$z = \rho\, z_{graph} + (1 - \rho)\, z_{tabular},\ \rho = \sigma(a) \quad (1)$$

where a is learned. A separate MLP uses the tabular branch alone. The selected graph and MLP candidates use ordinary empirical-risk minimization. A CORAL-inspired covariance alignment option [14], feature dropout, GroupDRO, and alternative graph masks are available but are not established improvements in this pilot.

An optional physics branch can compute symmetrical-component consistency features and a masked smooth-L1 auxiliary reconstruction objective. Enabling it requires explicit metadata verification and a reference. In the executed configuration, physics verification is false, auxiliary weights are zero, and this branch contributes no physical constraint. The model processes individual rows; no temporal sequence or BiLSTM is used.

E. Centralized Baselines and Ensemble

The comparison includes histogram gradient boosting and XGBoost with and without logs, ExtraTrees, MLP, and EdgeGT. Each family explores two candidate configurations. Tree fitting uses source-balanced and class-balanced sample weights; neural ERM uses class-balanced binary cross-entropy. Centralized neural fitting uses AdamW with cosine decay and gradient clipping. These objective and optimizer differences are important when interpreting model comparisons.

The ensemble combines the strongest inner-selected tree with EdgeGT. Eleven mixture weights from zero to one are evaluated on training-source out-of-fold probabilities at threshold 0.5. The weight is selected without using test results. MLP is not an eligible ensemble member in this implementation; therefore, the ensemble does not exhaust combinations involving the observed leading family.

F. Federated Training Simulator

Training files are assigned intact to four simulated clients using greedy size balancing. Every client participates in each round. Local optimization uses constant-rate SGD and twenty randomly sampled minibatch updates per client. The sample-size-weighted FedAvg update is

$$\theta_{r+1} = \sum_k \frac{n_k}{N}\, \theta_{k,r+1},\ N = \sum_k n_k \quad (2)$$

where n_k is client k's row count. FedProx adds a quadratic penalty relative to the broadcast parameters. FedAdam applies bias-corrected server moments to the mean model delta. SCAFFOLD corrects local gradients and updates client controls using the constant-step Option-II displacement formula [12]. Its model aggregation is sample-weighted while server controls are averaged uniformly; this weighted variant needs a separate aggregation-consistency check.

For each algorithm, two optimizer settings are compared using the same three inner source folds. The pilot permits up to forty rounds, with final refitting at the selected median round count. Source-based partitions are executed; IID and Dirichlet label partitions are implemented options but are not measured here. The benchmark runs in one process. It simulates local optimization and aggregation without secure aggregation, differential privacy, network transport, or distinct utility ownership.

G. Calibration and Decision Threshold

Training-source out-of-fold probabilities determine a positive temperature T by minimizing source-weighted log loss. Calibration then maps a raw attack probability p to

$$p_T = \sigma\left(\frac{logit(p)}{T}\right) \quad (3)$$

A grid of thresholds from 0.02 to 0.98 maximizes training-source macro-F1 on the calibrated probabilities; ties prefer the value nearest 0.5. The fitted temperature and threshold are frozen for the outer test. This keeps test labels outside postprocessing. However, the candidate-selection criterion at 0.5 differs from the final threshold criterion, and monotone temperature scaling does not create better score ranking.

IV. Experimental Setup

A. Executed Configuration and Source Composition

TABLE I
EXECUTED PILOT CONFIGURATION

Setting	Value
Outer / inner protocol	5 configured outer folds; fold 0 run; 3 inner folds
Training / test rows	62,635 / 15,732
Training / test files	12 / 3
Split / training seeds	42 / 11
Batch size	512
Central epoch budget	30; patience 8
FL round / local-step budgets	40 / 20
FL clients / partition	4 / source
Physics features	Disabled
Bootstrap replicates	2,000; diagnostic only

Table I lists budget limits rather than selected refit durations. The metadata marks the pilot complete and the full protocol incomplete. CUDA is reported without an embedded hardware or timing record; deployment latency and measured bandwidth are therefore not claimed.

TABLE II
HELD OUT TEST FILE COMPOSITION

File	Natural	Attack	Total
data12.csv	1,771	3,453	5,224
data4.csv	1,800	3,402	5,202
data8.csv	1,536	3,770	5,306
Total	5,107	10,625	15,732

The test population in Table II is attack-heavy: 67.54% of rows have the positive label. The majority comparator predicts Attack for every test row. Its pooled accuracy is therefore 67.54%, while its MCC is zero. This motivates class-balanced metrics and explicit false-alarm reporting rather than accuracy-only model selection.

B. Evaluation Measures and Uncertainty

For each source file, class-macro F1 averages the F1 scores of Attack and Natural. The primary score weights source files equally:

$$F1_{source} = \frac{1}{|S|} \sum_s F1_{macro,s} \quad (4)$$

where S is the set of test filenames. Pooled macro-F1 instead aggregates all test rows before class averaging. The study also reports pooled MCC, AUROC, Brier loss, and FPR = FP/(FP + TN). AUROC assesses ranking, Brier evaluates probability quality, and FPR quantifies the proportion of nonattack rows incorrectly flagged. Brier alone cannot isolate calibration from discrimination.

The notebook bootstraps source-level F1 scores with 2,000 resamples. Since only three files and one training seed are evaluated, the resulting intervals are conditional diagnostics. They do not include fitting or split uncertainty. The full configuration fits three seeds and averages their probabilities; this ensemble estimate should be separated from future seed-wise summaries. Individual rows and training seeds should not be treated as independent acquisition groups.

V. Results and Discussion

A. Centralized and Federated Comparison

TABLE III
OUTER TEST PERFORMANCE FOR ONE PILOT FOLD

Model	Source F1	MCC	AUC	Brier	FPR (%)
Majority	0.4029	0.000	0.5000	0.3240	100.0
HistGB + logs	0.5978	0.196	0.6770	0.2129	55.3
HistGB − logs	0.5976	0.196	0.6763	0.2129	55.7
XGB + logs	0.6011	0.203	0.6841	0.2119	54.0
XGB − logs	0.5984	0.201	0.6796	0.2131	59.0
ExtraTrees	0.5726	0.150	0.6091	0.2219	63.6
MLP	0.6109	0.223	0.6837	0.2192	52.0
EdgeGT	0.5974	0.201	0.6729	0.2197	47.1
Ensemble	0.5932	0.189	0.6816	0.2137	51.2
FedAvg	0.5583	0.119	0.5767	0.2465	62.8
FedProx	0.5478	0.102	0.5559	0.2519	53.6
FedAdam	0.5287	0.060	0.5390	0.2503	59.3
SCAFFOLD	0.5235	0.050	0.5380	0.2508	69.4

In Table III, F1 is source-averaged; the remaining measures are pooled across rows. Fig. 2 displays the same F1 ranking. MLP achieves the highest source F1, 0.6109, and MCC, 0.2226. XGBoost with logs has the highest AUROC, 0.6841, and the lowest Brier loss, 0.2119. These different leaders show that threshold-dependent classification and probability ranking are distinct evaluation goals.

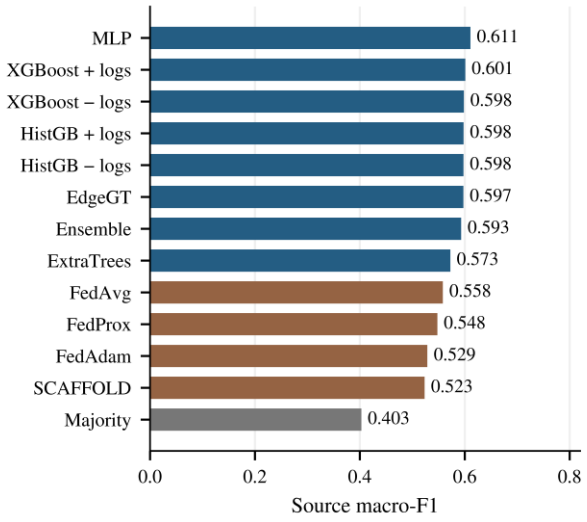


Fig. 2. Source macro-F1 from the saved v2 pilot. Blue bars denote centralized families, brown bars denote federated variants, and gray denotes the majority rule. No run-to-run uncertainty is implied by these single-fold bars.

EdgeGT achieves source F1 0.5974, below MLP by 1.35 percentage points and below XGBoost with logs by 0.37 points. The graph–tree ensemble scores 0.5932. This pilot therefore does not support a claim that graph structure or ensembling improves on the strongest centralized competitor. The learned tabular contribution and global-token connections further prevent attribution of any graph score solely to physical line adjacency.

FedAvg is the strongest federated variant, with source F1 0.5583 and AUROC 0.5767. Relative to centralized EdgeGT, the source-F1 gap is 3.91 points. FedProx, FedAdam, and SCAFFOLD score 0.5478, 0.5287, and 0.5235, respectively. Their lower ranking quality cannot be explained only by threshold choice. Because centralized AdamW and federated SGD have different optimization trajectories, these comparisons establish a pipeline-level gap rather than a controlled estimate of the cost of decentralization.

B. Cyber Logs and Per Source Behavior

Including logs changes HistGB source F1 from 0.5976 to 0.5978 and XGBoost from 0.5984 to 0.6011. The respective gains are 0.02 and 0.27 percentage points. These are small descriptive changes and do not establish a statistically reliable log contribution. Since preprocessing and sample weighting are shared within each pair, the comparison is nevertheless useful for planning a dedicated feature-group ablation.

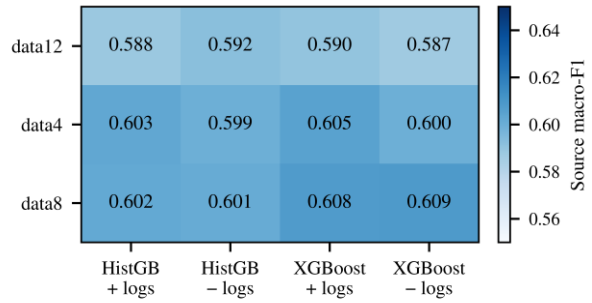


Fig. 3. Per-source F1 for four tree configurations, redrawn from the three saved source tables. The displayed color range is 0.55–0.65; cell labels provide exact values rounded to three decimals.

Fig. 3 shows modest tree performance across every held-out file. For XGBoost with logs, source F1 is 0.5902 on data12, 0.6053 on data4, and 0.6078 on data8. Thus, its overall limitation is not caused by one isolated failure. Similar source and pooled F1 values largely reflect the comparable test-file sizes; they are not interchangeable definitions.

C. Error Structure and Operational Thresholds

Among trained models, FPR ranges from 47.09% for EdgeGT to 69.43% for SCAFFOLD. MLP and XGBoost with logs have FPRs of 52.01% and 53.98%. These rates describe false alarms among negative rows, not the fraction of alerts that are false. Even the lowest observed rate is unsuitable evidence for a low-false-alarm deployment claim.

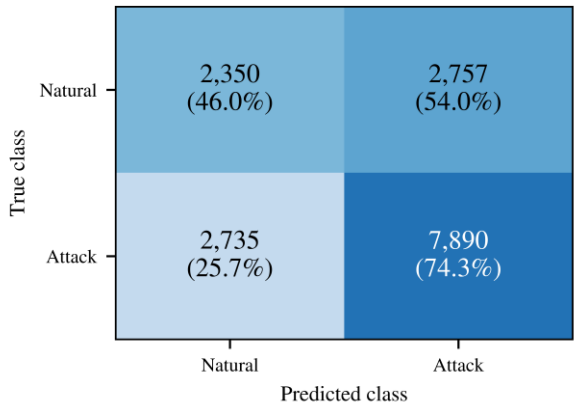


Fig. 4. XGBoost with logs: pooled confusion matrix obtained by summing the three saved source matrices. Counts are exact saved counts; percentages are normalized within each true class. Attack is the positive class.

The exact source counts yield the confusion matrix in Fig. 4: TN = 2,350, FP = 2,757, FN = 2,735, and TP = 7,890. Attack recall is 74.26%, and Natural recall is 46.02%. Its pooled accuracy is 65.09%, below the majority comparator, but its nonzero negative-class recognition raises pooled macro-F1 to 0.6015. The comparison illustrates why majority accuracy must be accompanied by both class recalls and balanced measures.

Deployment-oriented analysis should select operating thresholds on training-source predictions and measure test recall at prespecified low-FPR targets. Candidate targets such as 1% and 5% FPR are useful diagnostics, not guaranteed feasible outcomes. The current source-F1-optimized thresholds favor a different trade-off. Detailed ROC or PR trajectories cannot be reconstructed from a single reported AUC and operating point; they require prediction-level scores.

D. Probability Quality and Source Bootstrap Diagnostics

The majority comparator assigns probability 0.999 to its predicted class, yielding Brier loss 0.3240. It is a hard-label baseline rather than a fair constant-probability predictor. At the held-out prevalence, the best constant probability has Brier $p(1 - p) = 0.21924$. This test-prevalence value is an oracle diagnostic, not an admissible training baseline. A proper deployable comparator must estimate prevalence exclusively from training data.

MLP's Brier value of 0.21916 is close to that diagnostic constant, while EdgeGT scores 0.21972. XGBoost with logs yields 0.21186. Federated values range from 0.24653 to 0.25195, indicating weaker probability prediction under this evaluation. Reliability diagrams and raw-versus-calibrated NLL are needed to determine whether calibration, score resolution, or both drive these differences.

TABLE IV
DIAGNOSTIC SOURCE BOOTSTRAP INTERVALS

Model	Source F1	95% interval
MLP	0.6109	[0.6027, 0.6207]
XGB + logs	0.6011	[0.5902, 0.6078]
EdgeGT	0.5974	[0.5798, 0.6199]
FedAvg	0.5583	[0.5396, 0.5852]
SCAFFOLD	0.5235	[0.4861, 0.5427]

Table IV summarizes the saved intervals, each based on the same three files. Their precision should not be interpreted as evidence from thousands of independent sources: additional bootstrap draws only resample those three observed groups. Paired source-level comparisons and complete source coverage are preferable for final inference. Interval overlap or non-overlap by itself is not a significance test.

VI. Limitations and Future Work

A. Dataset Provenance and Scope of Generalization

The groups are filenames from a common testbed, not verified independent utilities, campaigns, or acquisition periods. Exact-record filtering cannot establish session independence. This one-fold, one-seed evaluation does not verify architecture superiority or physical constraints. Stronger claims require scenario/session groups, complete repeated evaluation, and an independent acquisition dataset. Choices informed by this pilot should be frozen before confirmation outcomes are inspected.

B. Preprocessing and Selection Alignment

A native-NaN, unclipped tree branch should test sensitivity to shared preprocessing. Audit clipping, missingness, finite sentinels, and training-only channel variation. Periodic angular encoding requires verified units and channel meaning; unknown sentinels should not be assigned an assumed physical interpretation.

Inner fitting and outer refitting use different cosine horizons. Separate schedule and stopping horizons to preserve the learning-rate trajectory. Align candidate selection, ensemble weighting, and final threshold objectives within training data, and use additional cross-fitting to examine selection effects in calibration.

C. Architecture and Federation Ablations

Compare graph-only, tabular-only, mixed, zero-edge-bias, line-only, alternative-topology, and full-attention variants. Report the learned mixture. Extend prespecified OOF ensembles to MLP and single-member endpoints. Efficiency claims require parameter counts, measured throughput, and memory.

Add FL-MLP and pooled centralized SGD with matched data exposure. Tune local steps, learning rate, class weighting, and aggregation before expanding rounds. Check consistent SCAFFOLD objectives. Its counted model/control payload doubles at equal rounds, model size, and participation, but excludes network overhead and is not measured bandwidth.

Complete five outer folds; report seed-wise and ensemble metrics separately; compare source, IID, and Dirichlet partitions. Test verified physics features independently. Privacy and adversarial robustness require explicit mechanisms and threat-model experiments.

VII. Conclusion

Fed-PIEdgeGT v2 integrates grouped selection, training-only preparation, OOF postprocessing, and source-aware reporting. In the pilot, MLP leads source F1 at 0.6109 and XGBoost leads AUROC at 0.6841. EdgeGT and FedAvg achieve source F1 of 0.5974 and 0.5583. Graph and federated variants do not outperform the strongest centralized baselines, and false-positive rates remain high. These outcomes motivate preprocessing checks, matched optimizer comparisons, representation ablations, and complete repeated evaluation before physics-informed superiority or deployment claims.

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